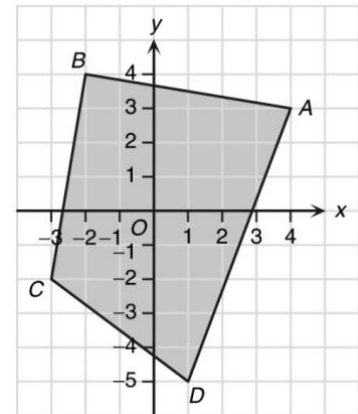


A001)

Teaching Example 10.1

Consider the rectangular coordinate plane as shown.

- (a) Write down the coordinates of points A and C . Which quadrants do they lie in?
- (b) What is the x -coordinate of point B ?
- (c) What is the y -coordinate of point D ?



A002)

Teaching Example 10.2

- (a) Plot four points $A(-4, 1)$, $B(-1, -2)$, $C(6, 1)$ and $D(3, 4)$ on a rectangular coordinate plane.
- (b) Join AB , BC , CD and DA .
- (c) Find the coordinates of the point of intersection of the diagonals of $ABCD$.

A003)

Teaching Example 10.3

Which of the following points is further away from $A(1, -8)$?

$P(-2, -8)$, $Q(3, -8)$

A004)

Teaching Example 10.4

Which of the following points is further away from $A(-3.5, -6.5)$?

$P(-3.5, -16.5)$, $Q(-3.5, 6.5)$

A005)

Teaching Example 10.5

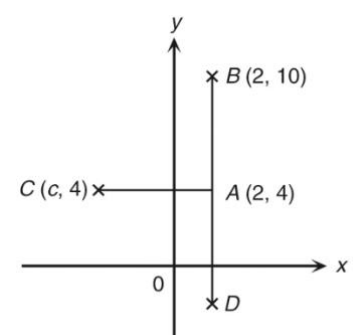
The four vertices of a rectangle are located at $P(4, 9)$, $Q(-1, 9)$, $R(-1, -7)$ and $S(4, -7)$. Find the perimeter of rectangle $PQRS$.

A006)

Teaching Example 10.6

In the figure, BAD is a straight line. Points B , C and D are of the same distance from A .

- (a) Find the distance between points A and B .
- (b) Find the value of c .
- (c) Find the coordinates of point D .

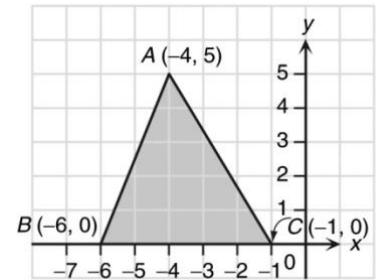


A007)

Teaching Example 10.7

The figure shows a triangle with vertices $A(-4, 5)$, $B(-6, 0)$ and $C(-1, 0)$.

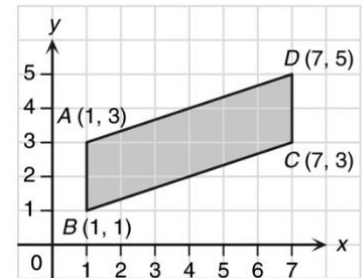
Find the area of $\triangle ABC$.



A008)

Teaching Example 10.8

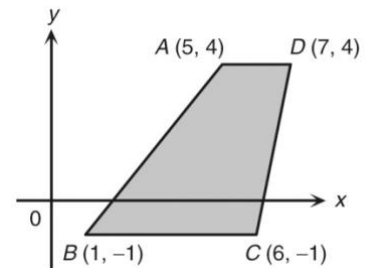
Find the area of parallelogram $ABCD$ in the figure.



A009)

Teaching Example 10.9

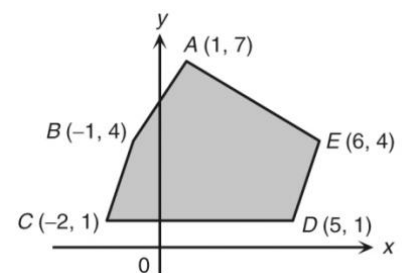
Find the area of trapezium $ABCD$ in the figure.



A010)

Teaching Example 10.10

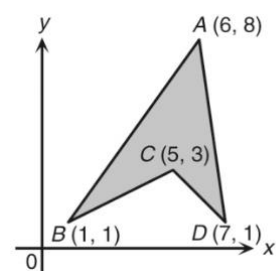
Find the area of pentagon $ABCDE$ in the figure.



A011)

Teaching Example 10.11

Find the area of quadrilateral $ABCD$ in the figure.



A012)

Teaching Example 10.12

$P(5, 8)$ is a point on a rectangular coordinate plane.

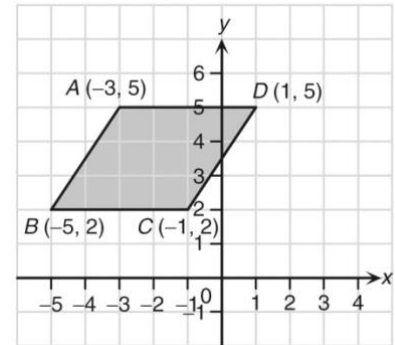
- (a) If P is translated to the right by 3 units to P_1 , find the coordinates of P_1 .
- (b) If P_1 is translated downwards by 1 unit to P_2 , find the coordinates of P_2 .
- (c) If P_2 is translated to the left by 6 units to P_3 , find the coordinates of P_3 .

A013)

Teaching Example 10.13

The figure shows parallelogram $ABCD$ with vertices $A(-3, 5)$, $B(-5, 2)$, $C(-1, 2)$ and $D(1, 5)$.

- (a) If $ABCD$ is translated to the right by 4 units and then translated downwards by 3 units to $A_1B_1C_1D_1$, draw $A_1B_1C_1D_1$ in the figure.
- (b) Find the coordinates of the vertices of $A_1B_1C_1D_1$.



A014)

Teaching Example 10.14

$P(1, 7)$ is a point on a rectangular coordinate plane. P is reflected about the x -axis to P_1 , and then P_1 is reflected about the y -axis to P_2 . Plot P , P_1 and P_2 on the same plane and write down the coordinates of P_1 and P_2 .

A015)

Teaching Example 10.15

L is a line parallel to the x -axis. It intersects the y -axis at $(0, 3)$. $P(7, 10)$ is reflected about L to P' . Find the coordinates of P' .

A016)

Teaching Example 10.16

If $A(-6, -1)$ is rotated anti-clockwise about the origin through 90° to A' , find the coordinates of A' .

A017)

Teaching Example 10.17

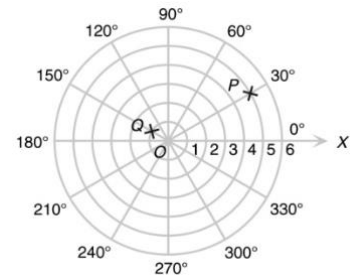
Consider a point $A(4, -9)$ on a rectangular coordinate plane.

- (a) If A is rotated anti-clockwise about the origin through 180° to B , find the coordinates of B .
- (b) If B is rotated anti-clockwise about the origin through 270° to C , find the coordinates of C .

A018)

Teaching Example 10.18

- (a) Write down the polar coordinates of P and Q on the following polar coordinate plane.
- (b) Plot $X(2, 180^\circ)$ and $Y(6, 210^\circ)$ on the above polar coordinate plane.
- (c) Find $\angle XOY$ and the length of PY .



A019)

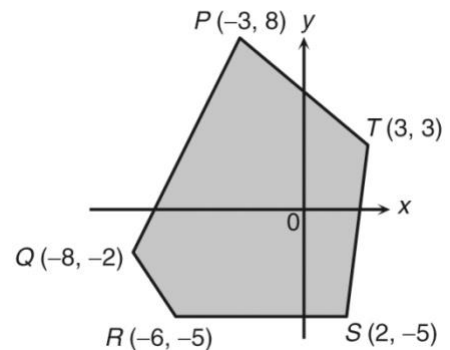
Teaching Example 10.6 (Extra)

Refer to teaching example 10.6. Find the coordinates of another point E , which is of the same distance from A as points B , C and D .

A020)

Teaching Example 10.11 (Extra)

Find the area of pentagon $PQRST$ in the figure.



A021)

Teaching Example 10.17 (Extra)

If $P(2, 9)$ is rotated clockwise about the origin through 90° , and then reflected about the y -axis to P' , find the coordinates of P' .

§10.2 Rectangular Coordinate System

D001)

The coordinates of A is $(2m + 1, m - 5)$. Find the coordinates of A if

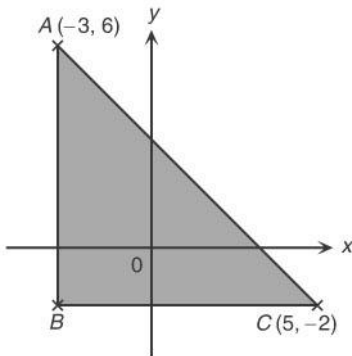
- (a) A lies on the x -axis,
- (b) A lies on the y -axis.

D002)

- (a) Plot $Q(-5, 10)$ on a rectangular coordinate plane and draw a line L passing through the origin O and Q .
- (b) Plot the following points on the plane and determine whether they lie on the line L .
- $(2, -1)$
 - $(3, -6)$

D003)

In the figure, $A(-3, 6)$ and $C(5, -2)$ are two vertices of $\triangle ABC$. AB is parallel to the y -axis and BC is parallel to the x -axis.

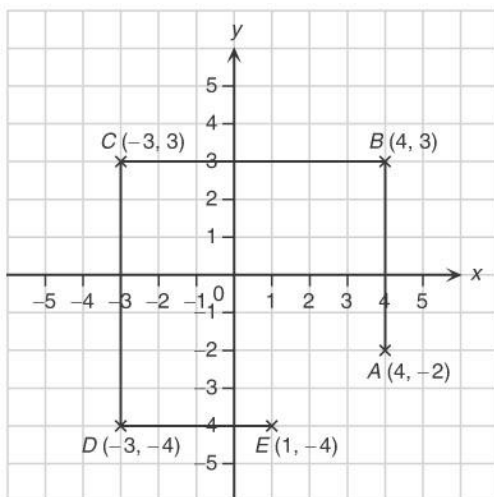


- (a) Write down the coordinates of B .
- (b) Determine whether the points $P(-3, 0)$, $Q(-1, -1)$ and $R(5, 1)$ lie inside, outside or on the side of $\triangle ABC$.

§10.3 Distance between Two Points

D004)

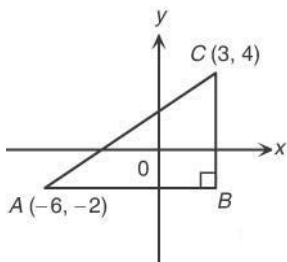
- (a) Find the lengths of line segments AB , BC , CD and DE in the figure.



- (b) Which line segment is the shortest?

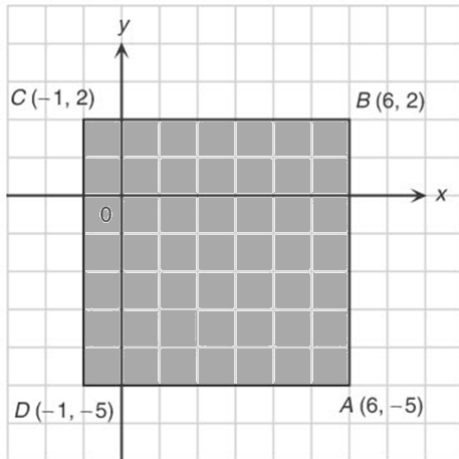
D005)

The figure shows a right-angled triangle ABC with $\angle B = 90^\circ$.



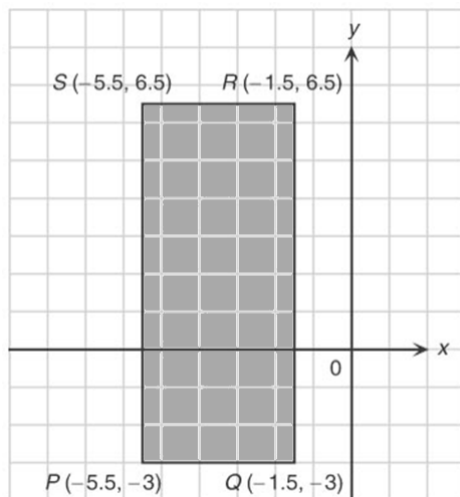
- (a) Find the coordinates of B .
- (b) Find AB and BC .

D006)

The figure shows a quadrilateral $ABCD$.

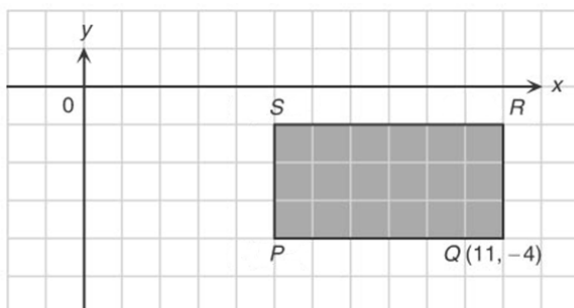
- (a) Find the lengths of AB , BC , CD and DA .
 (b) What kind of quadrilateral is $ABCD$?
 (c) Find the perimeter of quadrilateral $ABCD$.

D007)

The figure shows a quadrilateral $PQRS$.

- (a) Find the lengths of PQ , QR , RS and SP .
 (b) What kind of quadrilateral is $PQRS$?
 (c) Find the perimeter of quadrilateral $PQRS$.

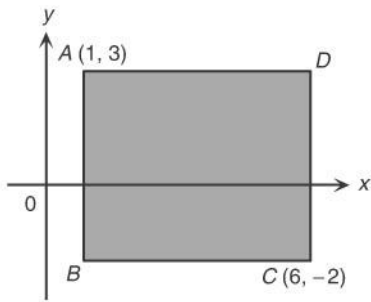
D008)

The figure shows a rectangle $PQRS$.

- (a) Write down the coordinates of P , R and S .
 (b) Find the perimeter of rectangle $PQRS$.

D009)

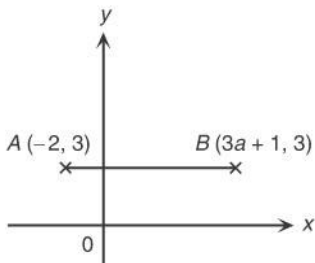
$A(1, 3)$ and $C(6, -2)$ are two vertices of rectangle $ABCD$ as shown in the figure. AB is parallel to the y -axis and BC is parallel to the x -axis.



- (a) Find the coordinates of B and D .
 (b) Find the perimeter of $ABCD$.

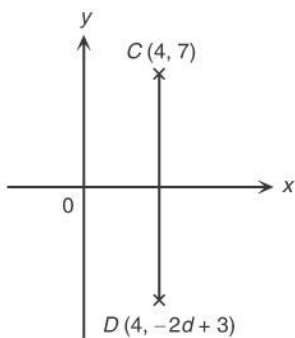
D010)

In the figure, $A(-2, 3)$ is on the left of $B(3a + 1, 3)$. If $AB = 9$ units, find a .



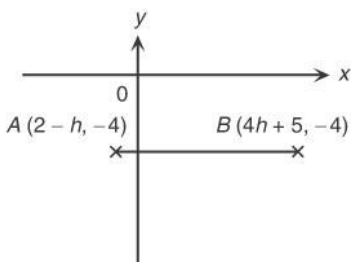
D011)

In the figure, $C(4, 7)$ is vertically above $D(4, -2d + 3)$. If $CD = 14$ units, find d .



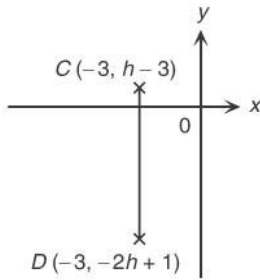
D012)

In the figure, $A(2 - h, -4)$ is on the left of $B(4h + 5, -4)$. If $AB = 18$ units, find h .



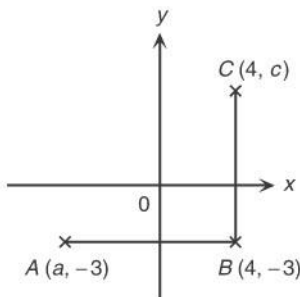
D013)

In the figure, $C(-3, h-3)$ is vertically above $D(-3, -2h+1)$. If $CD = 8$ units, find h .



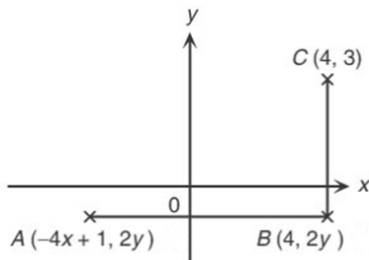
D014)

There are three points $A(a, -3)$, $B(4, -3)$ and $C(4, c)$ in the figure. If $AB = 9$ units and $BC = 8$ units, find a and c .



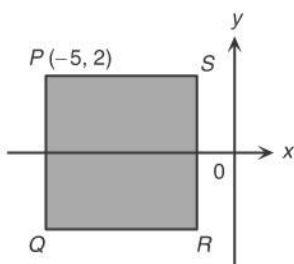
D015)

There are three points $A(-4x+1, 2y)$, $B(4, 2y)$ and $C(4, 3)$ in the figure. B lies on the right of A and C lies vertically above B . If $AB = 7$ units and $BC = 5$ units, find x and y .



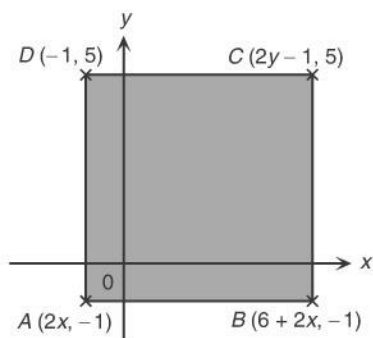
D016)

In the figure, $P(-5, 2)$ is a vertex of square $PQRS$. Given that the perimeter of $PQRS$ is 16 units, find the coordinates of Q , R and S .



D017)

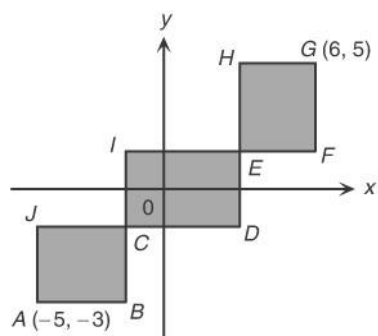
In the figure, the four vertices of square $ABCD$ are $A(2x, -1)$, $B(6 + 2x, -1)$, $C(2y - 1, 5)$ and $D(-1, 5)$.



- (a) Find AB .
 (b) Find x and y .

D018)

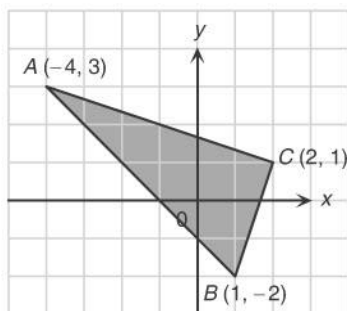
In the figure, AB , CD , EF , HG , IE and JC are parallel to the x -axis. JA , CB , ED , GF , HE and IC are parallel to the y -axis. If the coordinates of A and G are $(-5, -3)$ and $(6, 5)$ respectively, find the perimeter of the figure.



§10.4 Areas of Plane Figures

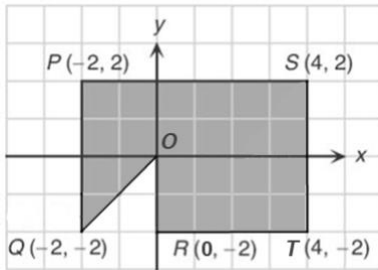
D019)

Find the area of $\triangle ABC$ in the figure.



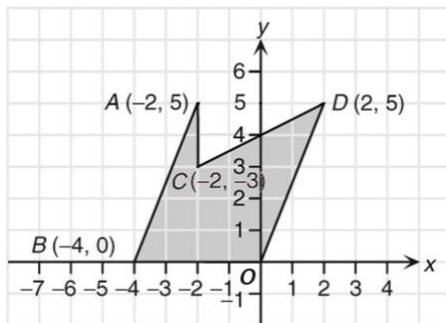
D020)

Find the area of the following figure.

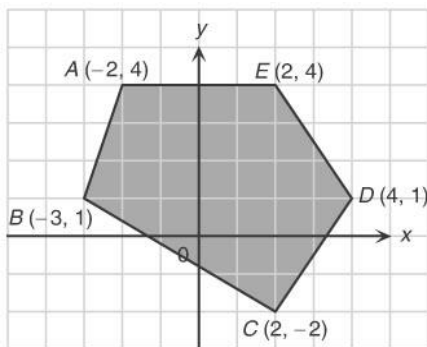


D021)

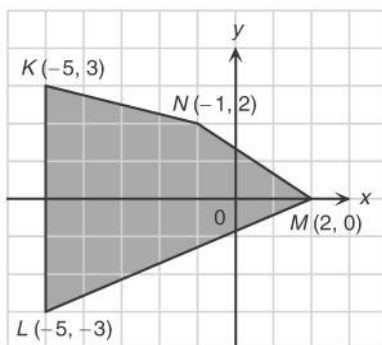
Find the area of the following figure.



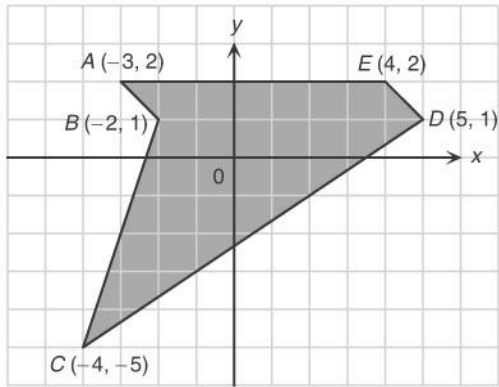
D022)

Find the area of pentagon $ABCDE$ in the figure.

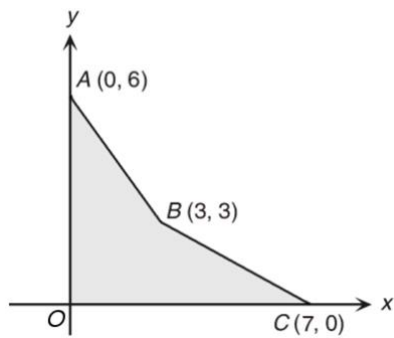
D023)

Find the area of quadrilateral $KLMN$ in the figure.

D024)

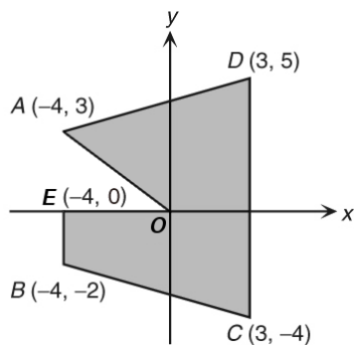
Find the area of pentagon $ABCDE$ in the figure.

D025)

Find the area of quadrilateral $OABC$ in the figure.

D026)

Find the area of the following figure.



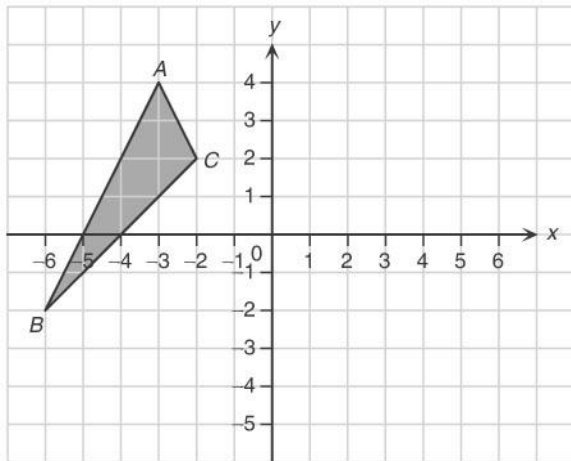
D027)

$ABCD$ is a parallelogram on a rectangular coordinate plane with vertices $A(3, -2)$, $B(1, 4)$ and $C(-4, 4)$. D lies in quadrant III.

(a) Find the coordinates of D .(b) Find the area of parallelogram $ABCD$.

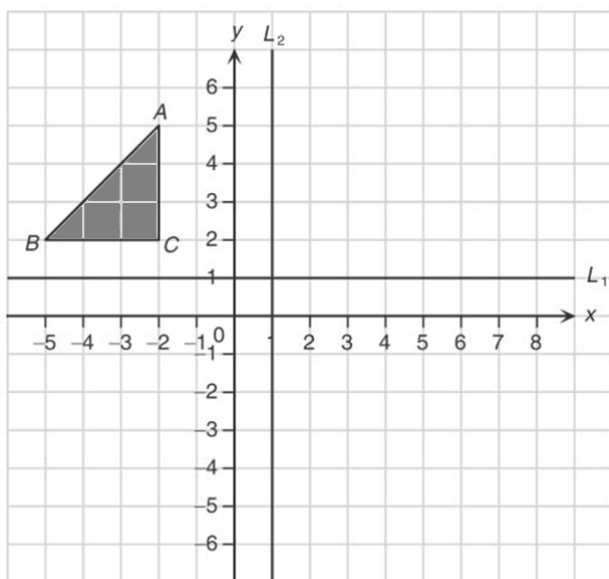
§10.5 Transformations of Points on the Coordinate Plane

D028)

The figure shows $\triangle ABC$.

- (a) If $\triangle ABC$ is translated downwards by 3 units to $\triangle A_1B_1C_1$, draw $\triangle A_1B_1C_1$ in the figure and write down the coordinates of the vertices of $\triangle A_1B_1C_1$.
- (b) If $\triangle A_1B_1C_1$ is reflected about the y -axis to $\triangle A_2B_2C_2$, draw $\triangle A_2B_2C_2$ in the figure and write down the coordinates of the vertices of $\triangle A_2B_2C_2$.

D029)

The figure shows $\triangle ABC$.

- (a) If $\triangle ABC$ is reflected about L_1 to $\triangle A_1B_1C_1$, draw $\triangle A_1B_1C_1$ in the figure and write down the coordinates of the vertices of $\triangle A_1B_1C_1$.
- (b) If $\triangle A_1B_1C_1$ is reflected about L_2 to $\triangle A_2B_2C_2$, draw $\triangle A_2B_2C_2$ in the figure and write down the coordinates of the vertices of $\triangle A_2B_2C_2$.

D030)

 $A(x + 2, 3)$ is reflected about the x -axis to $B(-2, y - 1)$.

- (a) Find x and y .
- (b) Write down the coordinates of A and B .

D031)

$P(3, 2y + 1)$ is reflected about the y -axis $Q(-x + 2, 5)$.

- (a) Find x and y .
 (b) Write down the coordinates of P and Q .

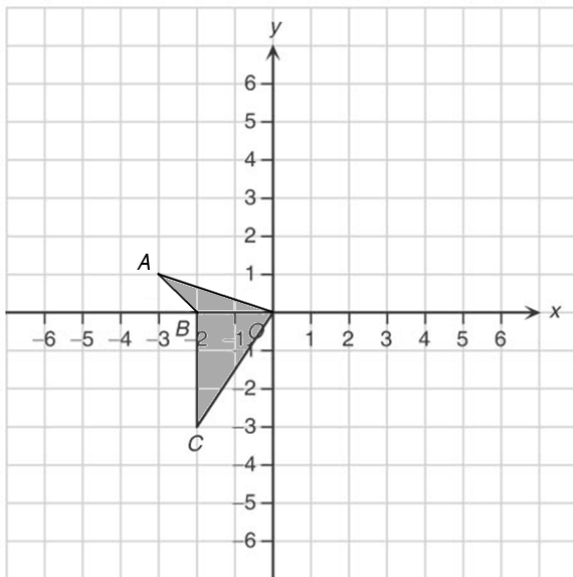
D032)

$M(1, 2 - y)$ is translated to the left by 5 units to $N(3 + x, 5)$.

- (a) Find x and y .
 (b) Write down the coordinates of M and N .

D033)

The figure shows a quadrilateral $OABC$.



- (a) It is given that $OABC$ is translated upwards by 5 units to $O_1A_1B_1C_1$. Draw $O_1A_1B_1C_1$ in the figure and write down the coordinates of O_1 , A_1 , B_1 and C_1 .
 (b) It is given that $O_1A_1B_1C_1$ is rotated clockwise about O through 90° to $O_2A_2B_2C_2$. Draw $O_2A_2B_2C_2$ in the figure and write down the coordinates of O_2 , A_2 , B_2 and C_2 .
 (c) If $O_2A_2B_2C_2$ is reflected about the x -axis to $O_3A_3B_3C_3$, draw $O_3A_3B_3C_3$ in the figure and write down the coordinates of O_3 , A_3 , B_3 and C_3 .

D034)

Given that $A(0, 5)$ is rotated anti-clockwise about the origin O through 90° and then translated upwards by 3 units to B ,

- (a) write down the coordinates of B ,
 (b) find the area of $\triangle OAB$.

D035)

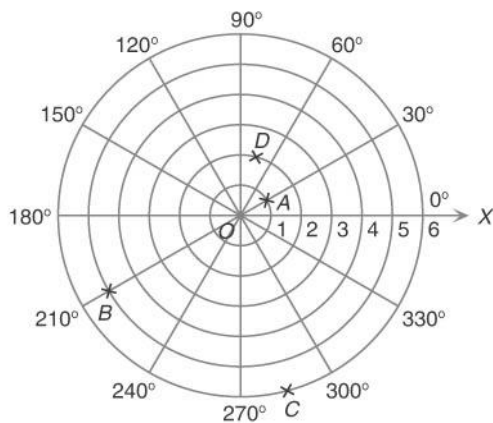
Given that $M(1, 3y + 1)$ is rotated anti-clockwise about the origin through 180° to $N(-x + 2, 2)$,

- (a) find x and y ,
 (b) write down the coordinates of M and N .

§10.6 Polar Coordinate System

D036)

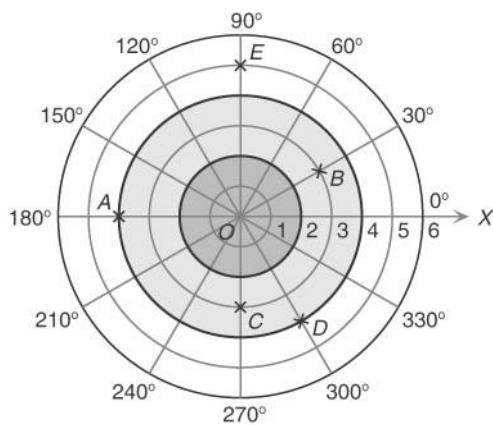
Refer to the following polar coordinate plane.



- Write down the polar coordinates of points A to D on the polar coordinate plane.
- Plot $E(3, 225^\circ)$ and $F(3, 315^\circ)$ on the figure.
- Plot G and H on the figure such that $EFGH$ is a square. Write down the polar coordinates of G and H .

D037)

Helen and Eva are playing an archery game. Each of them shoots five arrows.



- Helen hits A , B , C , D and E . Describe the positions of these points by using polar coordinates.
- Eva hits $F(1, 15^\circ)$, $G(3, 225^\circ)$, $H(2, 105^\circ)$, $I(2, 300^\circ)$ and $J(3, 120^\circ)$. Plot these points on the polar coordinates plane.
- If the one whose arrows attain a smaller total distance from O will be the winner, who will be the winner, Helen or Eva?

D038)

$P(4, 60^\circ)$ and Q are two points on a polar coordinate plane. If $OQ = 5$ units and $\angle POQ = 90^\circ$, where O is the pole of the polar coordinate plane,

- write down all the possible coordinates of Q ,
- find the area of $\triangle POQ$.

Level 3 Questions

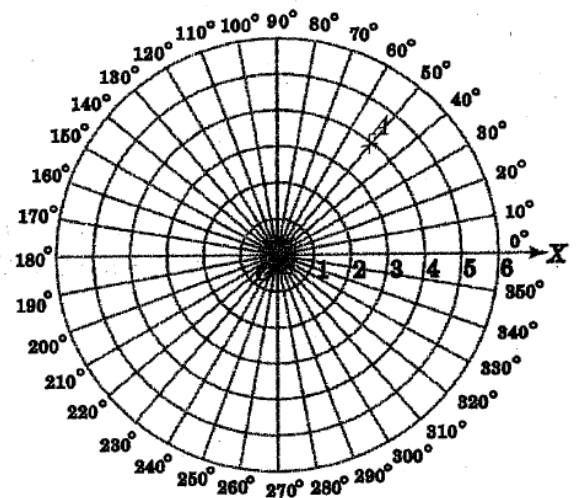
E001)

(百, 1617)

Refer to the polar coordinate plane.

(7 marks)

- Write down the polar coordinates of A .
- Plot $C(4, 230^\circ)$, $D(4, 320^\circ)$ on the figure.
- Plot B on the figure such that $ABCD$ is a square.
Write down the polar coordinates of B .
- Find the area of square $ABCD$.



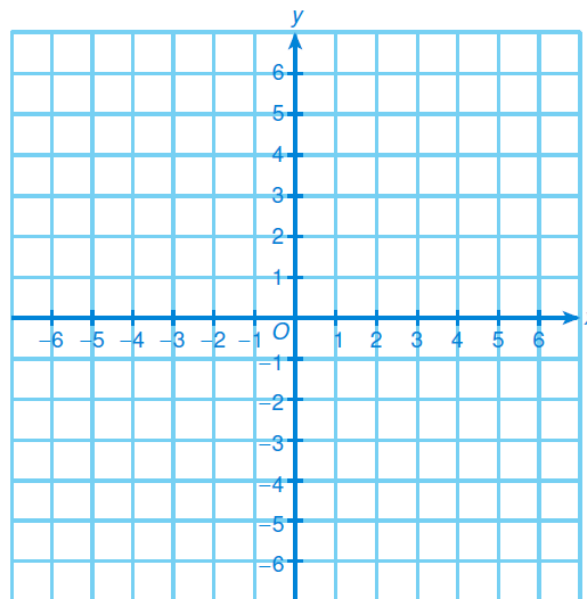
E002)

(梁, 1314)

Given three points $A(4, -5)$, $B(-4, -5)$ and $C(-2, 3)$.

- Mark the three points on the rectangular coordinate plane.

(1 mark)



- If $ABCD$ is a parallelogram, draw the parallelogram on the above coordinate plane and write down coordinates of D .
- If E is the intersection of AC and BD , write down the coordinates of E and which quadrant it lies in.
- Find the area of $\triangle BCE$.

(2 marks)

(2 marks)

(4 marks)

E003)

(陳, 1718)

In a polar coordinate system, O is the pole. The polar coordinates of the points P and Q are $(7, 120^\circ)$ and $(24, 210^\circ)$ respectively.

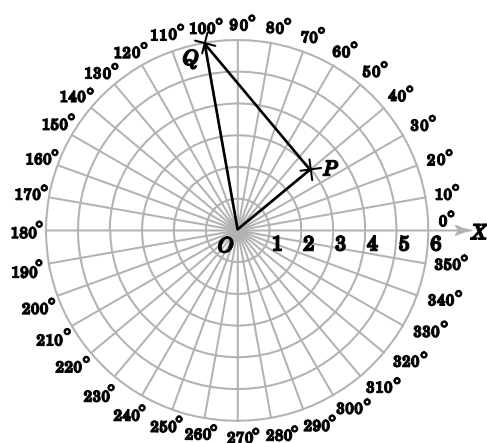
- Is $\triangle OPQ$ a right-angled triangle? Explain your answer.
- Find the area of $\triangle OPQ$.

(4 marks)

E004)

(呂, 1516)

The figure shows a polar coordinate plane.



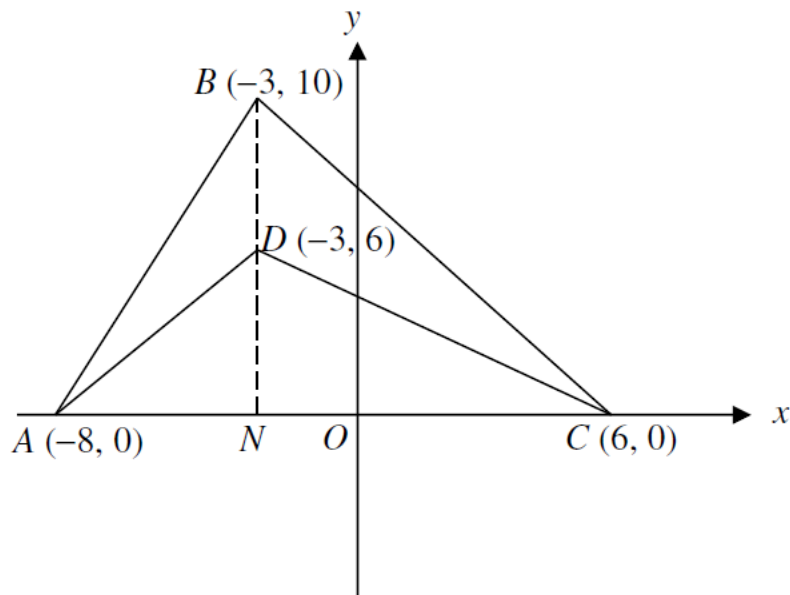
- Write down the polar coordinates of P and Q .
- Find $\angle POQ$.
- It is given that $\angle OQP = 30^\circ$. Andrew claims that $\triangle POQ$ is a right-angled triangle. Do you agree? Explain your answer.

(5 marks)

E005)

(劉, 1516)

In the figure, $ABCD$ is a quadrilateral. N is a point on the x -axis such that BN is perpendicular to the x -axis.



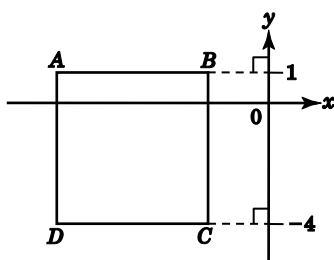
- (a) Find the coordinates of N . (1 mark)
- (b) Find the area of quadrilateral $ABCD$. (5 marks)

E006)

(呂, 1516)

The figure shows a square $ABCD$, where BC is a vertical line. Find the area of square $ABCD$.

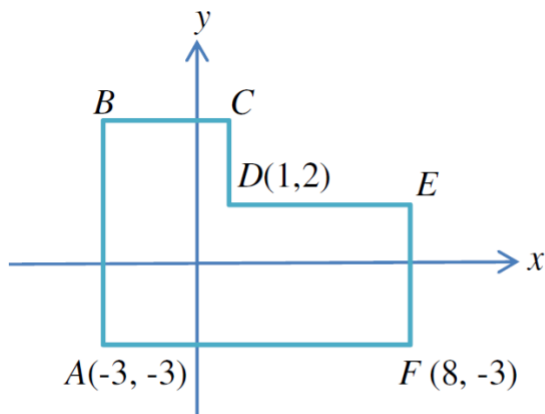
(4 marks)



E007)

(劉, 1314)

In the figure, AF , BC and DE are all parallel to the x -axis. AB , CD and EF are all parallel to the y -axis. The perimeter of $ABCDEF$ is 36 units.



- Find the length of AF and AB . (3 marks)
- Write down the coordinates of B , C and E . (4 marks)
- Find the area of $ABCDEF$. (3 marks)

E008)

(陳, 1718)

It is given that the coordinates of A and B are $(-5, 2)$ and $(10, 2)$ respectively. The area of rectangle $ABCD$ is 60 sq. units.

(a) Find AB . (2 marks)

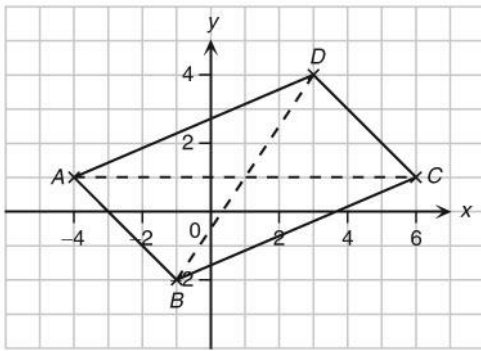
(b) (i) Find BC . (4 marks)

(ii) It is given that C and D are points above B . Hence, write down the coordinates of C and D .

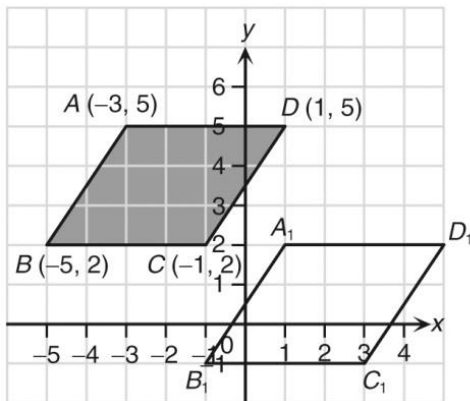
(c) If P is a point lying on the y -axis such that the area of $\triangle ABP$ is equal to the area of rectangle $ABCD$, write down all possible coordinates of P . (2 marks)

Answers

(1B10A001)(a) (4, 3), quadrant I, (-3, -2), quadrant III (b) -2 (c) -5 (1B10A002)(a), (b)

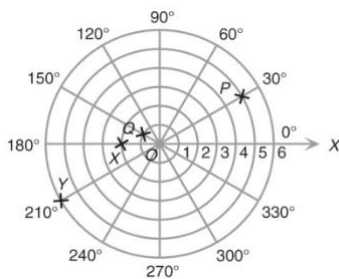


(c) (1, 1) (1B10A003) P (1B10A004) Q (1B10A005) 42 units (1B10A006)(a) 6 units (b) -4 (c) (2, -2) (1B10A007) 12.5 sq. units (1B10A008) 12 sq. units (1B10A009) 17.5 sq. units (1B10A010) 31.5 sq. units (1B10A011) 15 sq. units (1B10A012)(a) (8, 8) (b) (8, 7) (c) (2, 7) (1A10A013)(a)



(b) (1, 2), (-1, -1), (3, -1), (5, 2) (1B10A014) (1, -7), (-1, -7)

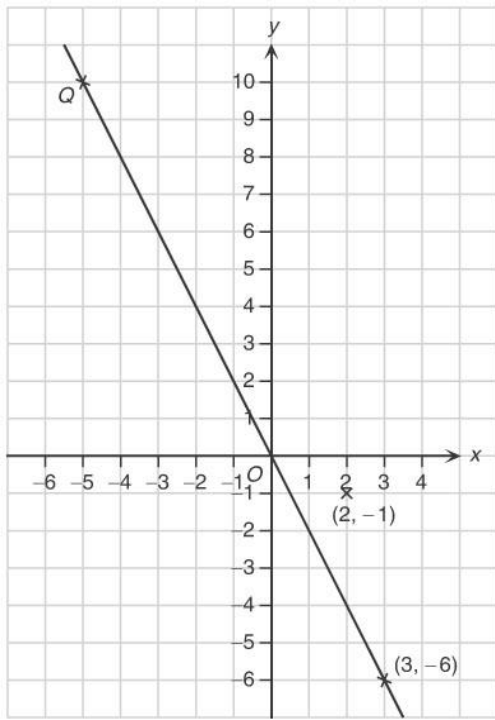
(1B10A015) (7, -4) (1B10A016) (1, -6) (1B10A017)(a) (-4, 9) (b) (9, 4) (1B10A018)(a) $P(5, 30^\circ)$, $Q(1, 150^\circ)$ (b)



(c) 11 units (1B10A019) (8, 4) (or any other reasonable answers)

(1B10A020) 96 sq. units (1B10A021) (-9, -2) (1B10D001)(a) (11, 0) (b) $(0, -5\frac{1}{2})$

(1B10D002)(a)



(b)(i) No (ii) Yes

(1B10D003)(a) $(-3, -2)$ (b) P : on the side, Q : inside, R : outside

(1B10D004)(a) AB = 5 units, BC = 7 units, CD = 7 unit, DE = 4 units (b) DE

(1B10D005)(a) $(3, -2)$ (b) AB = 9 units, BC = 6 units

(1B10D006)(a) AB = 7 units, BC = 7 units, CD = 7 units, DA = 7 units (b) square (c) 28 units

(1B10D007)(a) PQ = 4 units, QR = 9.5 units, RS = 4 units, SP = 9.5 units (b) rectangle (c) 27 units

(1B10D008)(a) P = $(5, -4)$, R = $(11, -1)$, S = $(5, -1)$ (b) 18 units(1B10D009)(a) B = $(1, -2)$, D = $(6, 3)$ (b) 20 units (1B10D010) 2 (1B10D011) 5 (1B10D012) 3(1B10D013) 4 (1B10D014) $-5, 5$ (1B10D015) 1, -1 (1B10D016) Q = $(-5, -2)$, R = $(-1, -2)$, P = $(-1, 2)$ (1B10D017)(a) 6 units (b) $y = 3, x = -\frac{1}{2}$

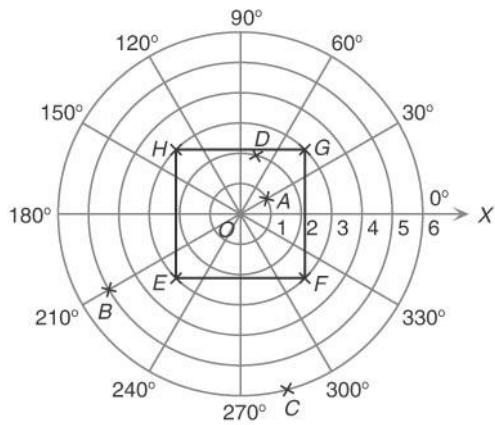
(1B10D018) 38 units

(1B10D019) 10 sq. units (1B10D020) 22 sq. units (1B10D021) 16 sq. units (1B10D022) 27 units

(1B10D023) 23.5 sq. units (1B10D024) 28 sq. units (1B10D025) 19.5 sq. units

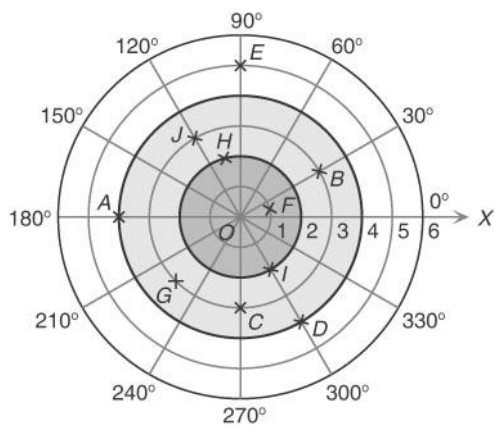
(1B10D026) 43 sq. units (1B10D027)(a) $(-2, -2)$ (b) 30 sq. units(1B10D028)(a) $A_1(-3, 1)$, $B_1(-6, -5)$, $C_1(-2, -1)$ (b) $A_2(3, 1)$, $B_2(6, -5)$, $C_2(2, -1)$ (1B10D029)(a) $A_1(-2, -3)$, $B_1(-5, 0)$, $C_1(-2, 0)$ (b) $A_2(4, -3)$, $B_2(7, 0)$, $C_2(4, 0)$ (1B10D030)(a) $-4, -2$ (b) $(-2, 3)$, $(-2, -3)$ (1B10D031)(a) 2 (b) $(3, 5)$, $(-3, 5)$ (1B10D032)(a) $-7, -3$ (b) $(1, 5)$, $(-4, 5)$ (1B10D033)(a) $O_1(0, 5)$, $A_1(-3, 6)$, $B_1(-2, 5)$, $C_1(-2, 2)$ (b) $O_2(5, 0)$, $A_2(6, 3)$, $B_2(5, 2)$, $C_2(2, 2)$ (c) $O_3(5, 0)$, $A_3(6, -3)$, $B_3(5, -2)$, $C_3(2, -2)$ (1B10D034)(a) $(-5, 3)$ (b) 12.5 sq. units(1B10D035)(a) 3, -1 (b) $(1, -2)$, $(-1, 2)$ (1B10D036)(a) $A(1, 30^\circ)$, $B(5, 210^\circ)$, $C(6, 285^\circ)$, $D(2, 75^\circ)$

(b)



(c) $G(3, 45^\circ)$, $H(3, 135^\circ)$ (1B10D037)(a) $A(4, 180^\circ)$, $B(3, 30^\circ)$, $C(3, 270^\circ)$, $D(4, 300^\circ)$, $E(5, 90^\circ)$

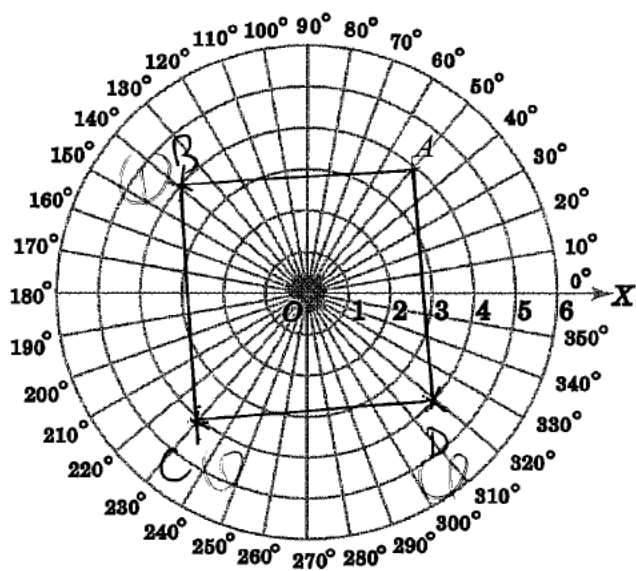
(b)



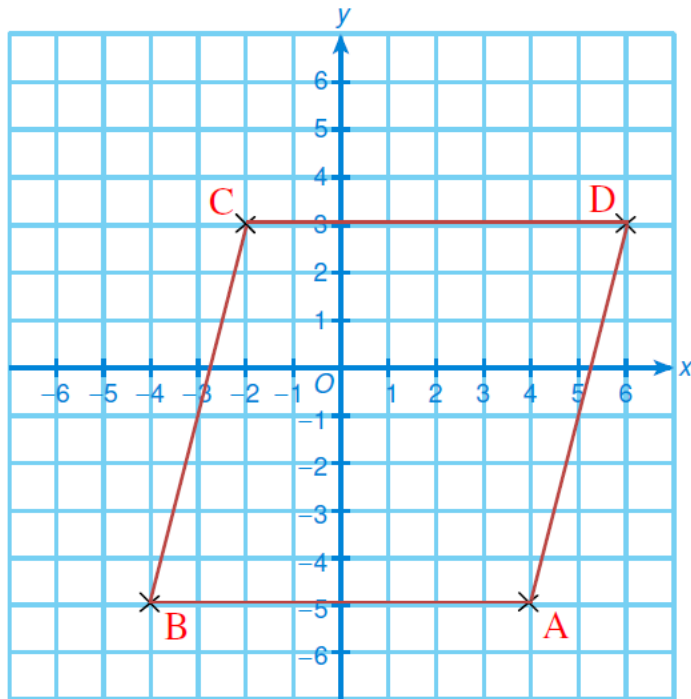
(c) Eva (1B10D038)(a) $(5, 150^\circ)$, $(5, 330^\circ)$ (b) 10 sq.

units (1B10E001)(a) $(4, 50^\circ)$

(b)



(c) $B = (4, 140^\circ)$ (d) 32 sq. units (1B10E002)(a)



(b) (6, 3) (c) (1, - 1), quadrant IV (e) 16 sq. units

(1B10E003)(a) yes (b) 84 sq. units

(1B10E004)(a) (3, 40°), (6, 100°) (b) 60° (c) yes

(1B10E005)(a) (- 3, 0) (b) 28

(1B10E006) 25 units

(1B10E007)(a) 11 units, 7 units (b) B = (- 3, 4), C = (1, 4), E (8, 1) (c) 63 sq. units

(1B10E008)(a) 15 units (b)(i) 4 units (ii) C = (10, 6), D = (- 5, 6) (c) (0, 10), (0, - 6)